Econ 202B: Lecture 2

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January 23, 2025

The goal for this lecture is to develop a basic model of households and firms that match patterns we see in the data.

1 Households

Households utility is given by

$$\mathbb{E}\left[\sum_{t=0}^{\infty}\beta^t\left(\ln C_t - \frac{L_t^{1+\frac{1}{\eta}}}{1+\frac{1}{\eta}}\right)\right]$$

We assume perfect foresight and we can ignore the expectation for now. Our representative agents seeks to maximize lifetime utility such that

$$A_t + C_t = (1+R_t)\,A_{t-1} + w_t L_t + \pi_t$$

Since we assume perfect competition, aggregate profits, $\pi_t = 0$. Assume that everyone is fully employed, so the distinction between hours and number employed does not mater. This distinction will matter if we were to relax this assumption. A_{t-1} are assets in the previous period that earn interest R_t.

We also assume a no Ponzi condition,

$$\lim_{t\to\infty}\left(\prod_{s=1}^{t}(1+R_s)\right)^{-1} A_s = 0$$

which means you can't die with debt.

The Lagrange is given by

$$\mathcal{L} = \sum \beta^t\left(\ln C_t - \frac{L_t^{1+\frac{1}{\eta}}}{1+\frac{1}{\eta}}\right) + \lambda_t\left(-A_t - C_t + (1+R_t)\,A_{t-1} + w_t L_t\right)$$

where the Lagrange multiplier, λ_t, is the marginal utility of wealth. Taking the FOCs, we get

$$\begin{aligned}
\frac{\partial\mathcal{L}}{\partial C_t} &= \frac{1}{C_t} - \lambda_t = 0 \Rightarrow \frac{1}{C_t} = \lambda_t \\
\frac{\partial\mathcal{L}}{\partial L_t} &= -L_t^{\frac{1}{\eta}} + \lambda_t w_t = 0 \Rightarrow L_t = (\lambda_t w_t)^{\eta}
\end{aligned}$$

This will give us an upward sloping labor supply curve. And finally for assets,

$$\begin{aligned}
\frac{\partial\mathcal{L}}{\partial A_t} &= -\lambda_t + \beta\,(1+R_{t+1})\,\lambda_{t+1} = 0 \\
&\Rightarrow \frac{1}{C_t} = \frac{1}{C_{t+1}}\beta\,(R_{t+1}+1)
\end{aligned}$$

which gives us the inter-temporal condition, or the Euler equation. This tells us how to spread our consumption over time.

If we assume

$$\bar{c} = c_t = c_{t+1}, \forall t$$

then

$$\begin{aligned} 1 &= \beta\left(1+\bar{R}\right) \\ &\Rightarrow \\ \bar{R} &= \frac{1}{\beta} - 1 \end{aligned}$$

When consumption is perfectly smooth, then the real interest rate is pinned down by our time preferences, β.

2 Firms

Firms own capital and seek to maximize cashflows. Present discounted cashflows are given by

$$\sum_{t=0}^{\infty}\left(\prod_{s=1}^{t}\left(1+R_s\right)\right)^{-1}\left(p_tY_t - w_tL_t - I_t\right)$$

Normalize $p_t = 1$ such that all prices are measured in terms of the final good prices. They optimize subject to

$$K_t = \left(1-\delta\right)K_{t-1} + I_t = K_{t-1} + \left(I_t - \delta K_{t-1}\right)$$

Production is given by

$$Y_t = Z_tK_{t-1}^{\alpha}L_t^{1-\alpha}$$

where Z_t is technology, K_{t-1}^{α} is pre-determined, and L_t is a free variable. Setting up the Lagrange, we get

$$\mathcal{L} = \sum_{t=0}^{\infty}\left(\prod_{s=1}^{t}\left(1+R_s\right)\right)^{-1}\left(Z_tK_{t-1}^{\alpha}L_t^{1-a} - w_tL_t - I_t + q_t\left(-K_t + \left(1-\delta\right)K_{t-1} + I_t\right)\right)$$

where q_t is Tobin's Q, and tells us whether we should divest (< 1) or invest (> 1). Taking the FOCs, we get

$$\begin{aligned} \frac{\partial\mathcal{L}}{\partial L_t} &= \left(1-\alpha\right)Z_tL_t^{-\alpha}K_{t-1}^{\alpha} - w_t = 0 \\ &\Rightarrow \\ w_t &= \left(1-\alpha\right)Z_t\left(\frac{K_{t-1}}{L_t}\right)^{\alpha} \\ &= \mathrm{MPL}\left(K_{t-1}, Z_t\right) \end{aligned}$$

This gives us our labor demand curve which are functions of the available capital stock and the productivity level.

$$\begin{aligned} \frac{\partial\mathcal{L}}{\partial I_t} &= -1 + q_t = 0 \Rightarrow q_t = 1 \\ \frac{\partial\mathcal{L}}{\partial K_t} &= -q_t + \frac{1}{1+R_{t+1}}\left(\alpha Z_{t+1}K_t^{\alpha-1}L_{t+1}^{1-\alpha} + q_t\left(1-\delta\right)\right) = 0 \\ &\Rightarrow \\ R_{t+1} + \delta &= \alpha Z_{t+1}\left(\frac{K_t}{L_{t+1}}\right)^{\alpha-1} \\ &= \mathrm{MPK}\left(L_{t+1}, Z_{t+1}\right) \end{aligned}$$

An important insight is that

$$\frac{\text{MPL}_t}{\text{MPK}_t} = \frac{w_t}{R_t + \delta} = \frac{1-\alpha}{\alpha}\left(\frac{K_{t-1}}{L_t}\right)$$

The interest rate and wage are pinned down by the level of technology and the capital-labor ratio. The relationship between wages and the interest rate is called the factor price possibility frontier (FPPF). A higher Z, means we will have to pay higher wages. Changes to the capital-labor ratio represent movements along the curve.

The labor supply curve is upward-sloping because of the negative term that appears in utility.

3 Market Clearing

There are three markets: labor, asset, and goods. By Walras Law, we can ignore the asset market. Nonetheless, the market clearing conditions for each are given by

$$\begin{aligned}
\text{Goods: } Y_t &= C_t + I_t \\
\text{Labor:} L^d &= L^s \\
\text{Asset:} A_t &= K_t
\end{aligned}$$

A market equilibrium is the sequence of prices $\{w_t, R_t\}_{t=0}^{\infty}$. Recall our production function,

$$Y_t = Z_t K_{t-1}^{\alpha} L_t^{1-\alpha}$$

The only source of variation in our model will come from productivity.

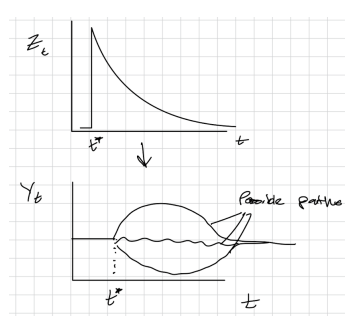
$$\begin{aligned}
Z_t &= Z_{t-1}^{\rho} \exp\left(\varepsilon_t\right) \\
&\Rightarrow \\
\ln Z_t &= \rho \ln Z_{t-1} + \varepsilon_t, \varepsilon_t \sim iid\left(0, \sigma_{\varepsilon}^2\right)
\end{aligned}$$

where ρ wants to be close to 0.

The system of equations that defines our equilibrium is

$$\begin{aligned}
C_t L_t^{\frac{1}{\eta}} &= (1-\alpha)\frac{Y_t}{L_t} \\
\frac{C_{t+1}}{C_t} &= \beta\left(\alpha\frac{Y_{t+1}}{K_t} + (1-\delta)\right) \\
Y_t &= Z_t K_{t-1}^{\alpha} L_t^{1-\alpha} \\
K_t &= (1-\delta) K_t + I_t \\
Z_t &= Z_{t-1}^{\rho} \exp\left(\varepsilon_t\right)
\end{aligned}$$

We want to see what happens to our system after a productivity shock. The evolution should look like,



4 Solving

Although the model is basic, the system is a set of non-linear equations which are difficult to solve. We could use the method of undetermined coefficients – which basically guessing and checking – to get a solution, or we could log-linearize.

We first need to make an assumption about the terminal conditions of our system. Assume that $\bar{Z} = 1$ in the steady state. The steady state variables are then $C_t = \bar{C}, L_t = \bar{L}, K_t = \bar{K}$. Our equilibrium in the steady state is thus,

$$\begin{aligned}
\bar{C}\bar{L}^{\frac{1}{\eta}} &= (1-\alpha)\frac{\bar{Y}}{\bar{L}} \\
1 &= \beta\left(\alpha\frac{\bar{Y}}{\bar{K}} + (1-\delta)\right) \\
\bar{Y} &= \bar{Z}\bar{K}^{\alpha}\bar{L}^{1-\alpha} \\
\bar{Y} &= \bar{C} + \bar{I} \\
\delta\bar{K} &= \bar{I}
\end{aligned}$$

We can take the logs of our production function at time t to get

$$\begin{aligned}
Y_t &= Z_t K_{t-1}^{\alpha} L_t^{1-\alpha} \\
&\Rightarrow \\
\ln Y_t &= \ln Z_t + \alpha \ln K_{t-1} + (1-\alpha)\ln L_t \\
&\Rightarrow \\
\ln\left(\frac{Y_t}{\bar{Y}}\right) &= \ln\left(\frac{Z_t}{\bar{Z}}\right) + \alpha\ln\left(\frac{K_{t-1}}{\bar{K}}\right) + (1-\alpha)\ln\left(\frac{L_t}{\bar{L}}\right)
\end{aligned}$$

which is an approximation of percent change for small changes. log-linearizing our entire system, we get

$$\begin{aligned}
\check{Y}_t &= \left(1 - \frac{\bar{I}}{\bar{Y}}\right)\check{C}_t + \frac{\bar{I}}{\bar{Y}}\check{I}_t \\
\check{K}_t &= (1-\delta)\check{K}_{t-1} + \delta\check{I}_t \\
\check{Z}_t &= \rho\check{Z}_{t-1} + \varepsilon_t \\
\check{C}_t + \left(\frac{1}{\eta}\right)\check{L}_t &= \check{Y}_t - \check{L}_t \\
\check{C}_{t+1} - \check{C}_t &= \frac{\alpha\left(\frac{\bar{Y}}{\bar{K}}\right)}{\alpha\left(\frac{\bar{Y}}{\bar{K}}\right) + (1-\delta)}\left(\check{Y}_{t+1} - \check{K}_t\right)
\end{aligned}$$

4.1 Calibration

Before we can test the model, there are several exogenous parameters we need to set before hand,

1. IES
2. Time preferences: β
3. Elasticity of output to L: $(1-\alpha)$
4. Persistence to technology shock: ρ
5. Volatility of technology shock: σ_{ε}^2
6. Fisch labor supply of elasticity: η
7. Depreciation rate: δ

When we evaluate the model, we need to make sure that the moments we are testing are orthogonal to the moments used to retrieve the parameters. Thus, it's common practice to retrieve these estimates from first moments of aggregate data and to test with their second moments

There have solutions proposed to the estimation:

1. By utility of consumption choice, $\ln C_t$, IES $= 1$.

2. By Euler equation, we can pin down the time preference by the interest rate which has remained relatively constant over time (Kaldor's facts).

$$\beta = \frac{1}{1+\bar{R}} \approx 0.96$$

3. We can use our equation for MPL o back out the elasticity of output to labor,

$$1-\alpha = \frac{w_t L_t}{Y_t} = \frac{\text{labor earnings}}{\text{GDP}} \approx \frac{2}{3} \text{ historically}$$

4. To estimate persistence, we can back out the Solow residual,

$$\check{Z}_t = \check{Y}_t - \alpha \check{K}_{t+1} + (1-\alpha)\check{L}_t$$

   where $SR_t = \check{Z}_t$ and run the following regression

$$SR_t = bSR_{t-1} + \text{error}$$

   so that b can be interpreted as ρ. There are measurement issues with this.

5. Macro wants the labor supply to be $\eta = \infty$, but micro tells us that $\eta = 0$. How does Macro deal with this? We argue that the low micro elasticities only account for the intensive margin. If we look at the unemployed pool, they are willing to work for no change to price level. We will settle with $\eta \approx 1$. This will become a huge problem for later where models rely on market clearing.

6. The useful life for building is approximately 100 years, so $\delta = 0.01$. The useful life for a PC is approximately 3 years, so $\delta = 0.3$. For cars its every 10 years, so $\delta = 0.1$. How do we measure the depreciation rate for something that doesn't have a resale market – nuclear submarines? There is clearly a lot of variation, selection, and measurement issues but the literature has decided to take on average on what we can observe and have settled on $\bar{\delta} \approx 0.1$.